

Evaluation of Radial Basis Function Networks on the MNIST Classification Task

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Abstract—This project examines the performance of a Radial Basis Function (RBF) neural network on the MNIST handwritten digit classification task and compares it with two classical non-parametric baselines: Nearest Neighbor (NN) and Nearest Class Centroid (NCC). The RBF model employs Gaussian basis functions, center selection via K-means clustering or random sampling, and a linear output layer trained using ridge regression. Principal Component Analysis (PCA) is applied to reduce dimensionality and investigate the trade-off between computational efficiency and classification accuracy.

A systematic experimental study is conducted by varying key hyperparameters, including the number of hidden neurons K , PCA variance retention, center initialization strategy, bandwidth estimation method, regularization strength, and random seeds. The results show that the RBF network achieves competitive performance across a wide range of configurations, reaching an accuracy of 96.98% for $K=1500$, slightly surpassing the 1-Nearest Neighbor baseline. More compact models (e.g., $K=200$) achieve accuracies around 94–95% with substantially lower training time, offering a favorable balance between accuracy and computational cost.

PCA is found to significantly reduce training time with minimal impact on accuracy, while K-means center selection consistently outperforms random initialization. Confusion matrices and example predictions highlight typical misclassification patterns among visually similar digits. Overall, the study provides a detailed analysis of how architectural and training choices influence the behavior of RBF networks and demonstrates their effectiveness as interpretable and flexible models for multi-class classification tasks.

I. INTRODUCTION

The classification of handwritten digits is a fundamental problem in machine learning, serving both as a benchmark for evaluating algorithms and as a gateway to understanding core concepts in pattern recognition. The MNIST dataset, consisting of grayscale images of digits written by different individuals, provides a widely adopted testbed for studying supervised learning methods due to its simplicity, accessibility, and well-established performance baselines.

Radial Basis Function (RBF) neural networks represent a classical yet powerful architecture for function approximation and classification. Unlike multilayer perceptrons, RBF

networks rely on localized activation functions centered around representative points in the input space, enabling them to model complex decision boundaries with interpretable geometric structure. Their performance depends critically on several design choices, including the selection of centers, the width of the radial basis functions, and the number of hidden units. Despite their conceptual simplicity, RBF networks remain relevant in modern machine learning due to their flexibility, fast inference, and strong theoretical foundations.

This project investigates the behavior of RBF networks on the MNIST digit classification task and compares their performance with two standard non-parametric baselines: Nearest Neighbor (NN) and Nearest Class Centroid (NCC). In addition, dimensionality reduction through Principal Component Analysis (PCA) is incorporated to study how feature compression affects accuracy and computational cost. A systematic experimental analysis is conducted by varying key hyperparameters such as the number of hidden neurons, PCA variance retention, center initialization strategy, bandwidth estimation method, regularization strength, and random seeds.

The goal of this study is to provide a comprehensive understanding of how architectural and training choices influence the performance of RBF networks, to evaluate their strengths and limitations relative to classical baselines, and to highlight the trade-offs between accuracy, interpretability, and computational efficiency. The results offer insight into the practical behavior of RBF models and demonstrate their capability to achieve competitive performance on a widely used multi-class classification benchmark.

II. DATASET AND PREPROCESSING

A. Dataset Description

The experiments in this study are conducted on the MNIST dataset, a widely used benchmark for image classification. MNIST consists of **70,000 grayscale images** of handwritten digits from 0 to 9, each of size **28×28 pixels**. The dataset is split into:

- **60,000 training samples**
- **10,000 test samples**

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Each image is represented as a flattened vector of **784 features**, corresponding to pixel intensity values normalized in the range [0,255].

B. Preprocessing

1. Normalization

All pixel values are scaled to the range [0,1] by dividing by 255. This normalization step improves numerical stability and ensures that distance-based methods (such as RBF networks and NN baselines) behave consistently.

2. Flattening

Each 28×28 image is reshaped into a 784-dimensional vector. No additional feature engineering or reshaping is applied, as the RBF model operates directly in the input space or in the PCA-reduced space.

3. Dimensionality Reduction with PCA

Principal Component Analysis (PCA) is applied to reduce the dimensionality of the input space before training the RBF network. PCA is fitted on the training set and then applied to both training and test data using the same transformation.

Three variance-retention levels are examined:

- **90% variance**
- **95% variance**
- **99% variance**

For the default configuration, retaining **95% variance** results in **154 principal components**, significantly reducing dimensionality while preserving most of the discriminative information. PCA is used to:

- reduce computational cost
- mitigate the curse of dimensionality
- improve the conditioning of the RBF design matrix

4. Train–Test Split

The standard MNIST split is used without modification:

- The **training set** is used for PCA fitting, center selection, and model training.
- The **test set** is used exclusively for evaluation.

No validation set is required, as the study focuses on controlled hyperparameter sweeps rather than model selection.

5. Summary of Preprocessing Pipeline

1. Load MNIST images and labels.

2. Normalize pixel intensities to [0,1].
3. Flatten images into 784-dimensional vectors.
4. Fit PCA on the training set and transform both sets.
5. Use the PCA-reduced data for all experiments.

III. CLASSIFIERS AND METHODOLOGY

This methodology enables a controlled and comprehensive comparison between simple non-parametric classifiers and a flexible RBF network. By varying architectural and training choices, the study highlights how each component influences accuracy, computational cost, and generalization.

Three classification methods are evaluated in this study:

1. **Nearest Neighbor (NN)**
2. **Nearest Class Centroid (NCC)**
3. **Radial Basis Function (RBF) Neural Network**

The first two serve as classical non-parametric baselines, while the RBF network constitutes the main model under investigation. All classifiers operate on the PCA-transformed feature space.

A. Nearest Neighbor (NN)

The Nearest Neighbor classifier assigns a test sample to the class of the closest training sample in Euclidean distance. Despite its simplicity, NN is known to perform strongly on MNIST due to the high density and diversity of the dataset.

Key characteristics:

- Non-parametric
- No training phase (lazy learner)
- High memory cost (stores all training samples)
- High inference cost (distance computation to all samples)

NN serves as an upper-bound baseline, as it typically achieves high accuracy on MNIST.

B. Nearest Class Centroid (NCC)

The NCC classifier represents each class by the mean vector of its training samples. A test sample is assigned to the class whose centroid is closest in Euclidean distance.

Key characteristics:

- Extremely fast training (computing 10 centroids)
- Very low memory footprint
- Lower accuracy than NN due to oversimplified class representation

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NCC provides a lightweight baseline that highlights the trade-off between simplicity and performance.

- Bandwidth estimation: median, per-center
- Ridge parameter $\lambda \in \{0.001, 0.01, 0.1\}$
- Random seeds {42, 123}

C. Radial Basis Function (RBF) Neural Network

The RBF network consists of three conceptual layers:

- **Input layer:** PCA-reduced feature vector
- **Hidden layer:** K radial basis functions
- **Output layer:** Linear classifier trained with ridge regression

1. Center Selection

Two strategies are examined:

- **K-means clustering:** centers correspond to cluster centroids
- **Random sampling:** centers selected uniformly from training samples

K-means typically yields more representative centers but increases training time.

2. Radial Basis Functions

Each hidden neuron computes:

Two bandwidth estimation methods are evaluated:

- **Global σ (median distance)**
- **Per-center σ (local neighborhood distances)**

3. Output Layer Training

The output weights are computed using ridge regression:

where:

- Φ is the design matrix of RBF activations
- Y is the one-hot encoded label matrix
- λ is the regularization parameter

4. Hyperparameters Explored

- Number of hidden neurons
 $K \in \{50, 100, 200, 400, 1500\}$
- PCA variance retention {90%, 95%, 99%}
- Center initialization: kmeans, random

D. Evaluation Methodology

All models are evaluated on the MNIST test set using:

- **Classification accuracy**
- **Training time** (for RBF and baselines)
- **Confusion matrices**
- **Qualitative inspection of correct and incorrect predictions**

The same PCA transformation is applied to both training and test data to ensure consistency. No hyperparameter tuning via validation sets is performed; instead, a systematic grid of experiments is executed to analyze model behavior.

IV. EXPERIMENTS AND RESULTS

All experiments are conducted on the MNIST test set using the PCA-transformed feature space. Unless otherwise stated, the default configuration uses:

- PCA variance retention: **95%** (154 components)
- Center initialization: **K-means**
- Bandwidth estimation: **median distance**
- Ridge parameter: $\lambda = \mathbf{0.01}$
- Random seed: **42**

Training time is measured for each model, and accuracy is computed on the 10,000-sample test set. All experiments are repeated across multiple hyperparameter settings to analyze the effect of architectural and training choices on performance.

A. Baseline Performance

Two classical non-parametric classifiers are used as baselines.

❖ Nearest Neighbor (NN)

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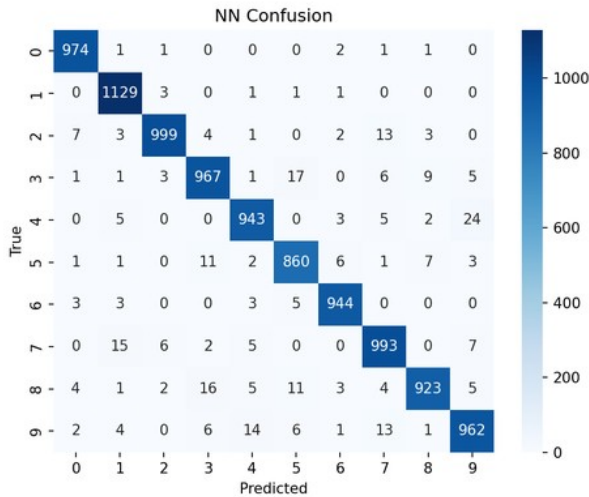


Figure 1: Confusion matrix for Nearest Neighbor

- Accuracy: **96.94%**
- Training time: ~2.4 s
- Strengths: high accuracy, simple, no training
- Weaknesses: slow inference, high memory cost

The NN classifier exhibits strong performance, with confusion patterns similar to those of the high-capacity RBF model. Remaining errors are largely due to intrinsic ambiguity in the MNIST dataset rather than model limitations.

❖ Nearest Class Centroid (NCC)

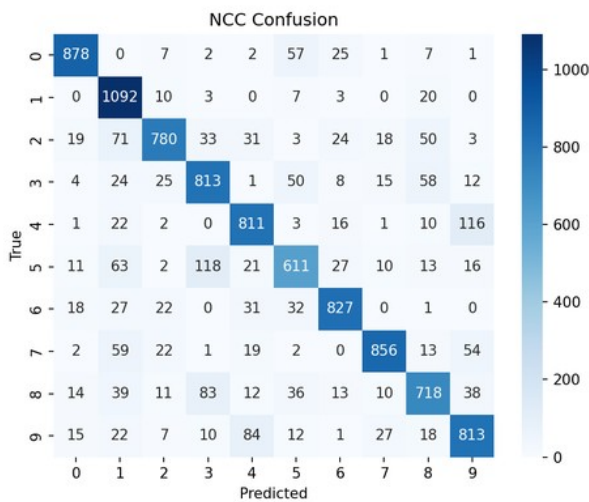


Figure 2: Confusion matrix for Nearest Class Centroid (NCC)

- Accuracy: **81.99%**
- Training time: ~0.09 s
- Strengths: extremely fast, minimal memory
- Weaknesses: low accuracy due to oversimplified class representation

The NCC classifier exhibits significantly lower accuracy compared to NN and RBF models, which is reflected in its confusion matrix. Many digits are misclassified across

multiple classes, indicating that representing each class with a single centroid is insufficient for capturing the variability of handwritten digits. Digits with high intra-class variability, such as 5, 8, and 9, show particularly high confusion rates. This highlights the limitations of NCC as an overly simplistic model for MNIST, despite its extremely low computational cost.

The baselines establish a performance range within which the RBF network can be evaluated.

B. RBF Baseline Configuration ($K = 200$)

Using the default configuration with $K = 200$ hidden neurons:

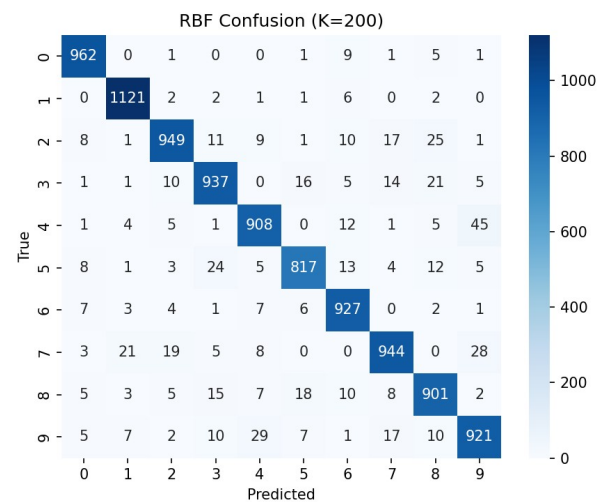


Figure 3: Confusion matrix for RBF ($K = 200$)

- Accuracy: **93.82%**
- Training time: ~60.7 s

This serves as the reference point for subsequent experiments. The confusion matrix reveals typical digit confusions such as:

- 4 ↔ 9
- 3 ↔ 5
- 8 ↔ 9

These patterns are consistent with known MNIST ambiguities.

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C. Effect of PCA Variance Retention

PCA variance levels of **90%, 95%, and 99%** are evaluated.

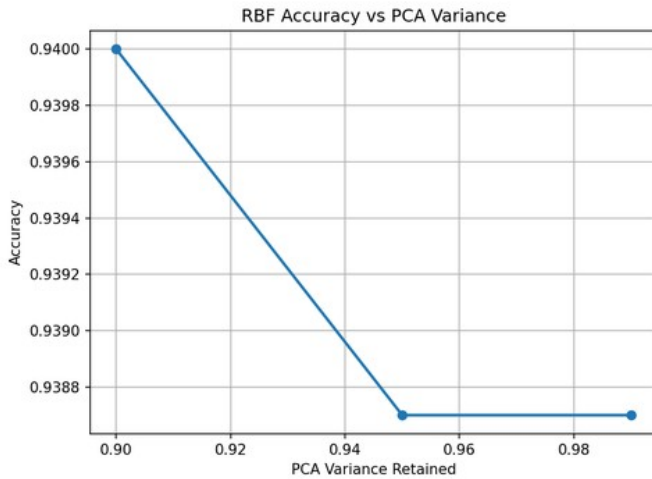


Figure 4: Test accuracy as a function of PCA variance retention

Interestingly, retaining more PCA variance does not improve accuracy. In fact, the model performs slightly better with 90% variance retention. This suggests that the additional components may introduce noise or redundant information that slightly degrades generalization. The result highlights that dimensionality reduction can act as a regularizer, improving performance by removing irrelevant variation. This indicates that the RBF network is robust to moderate dimensionality reduction.

D. Effect of Number of Hidden Neurons (K)

Values of $K \in \{50, 100, 200, 400, 1500\}$ are tested.

1. Accuracy vs K:

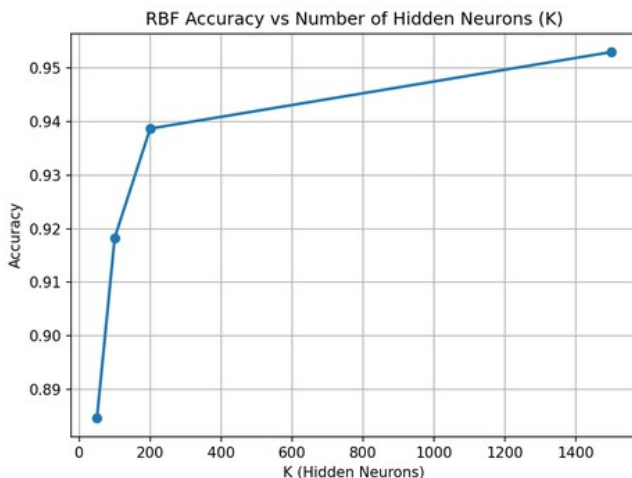


Figure 5: Test accuracy as a function of the number of RBF centers K

- $K = 50 \rightarrow 88.46\%$
- $K = 100 \rightarrow 91.83\%$
- $K = 200 \rightarrow 93.87\%$
- $K = 400 \rightarrow 95.30\%$
- **$K = 1500 \rightarrow 96.98\%$ (best)**

Increasing the number of RBF centers leads to a consistent improvement in classification accuracy. For small values of K (e.g., 50 or 100), the model lacks sufficient expressive power and underfits the data. Performance improves substantially for $K = 200$ and $K = 400$, indicating that additional centers allow the network to capture more complex decision boundaries. The highest accuracy is achieved at $K = 1500$, where the model reaches 96.98%, slightly surpassing the Nearest Neighbor baseline. This trend confirms the theoretical expectation that larger RBF networks can approximate more intricate class structures, albeit at a significantly higher computational cost.

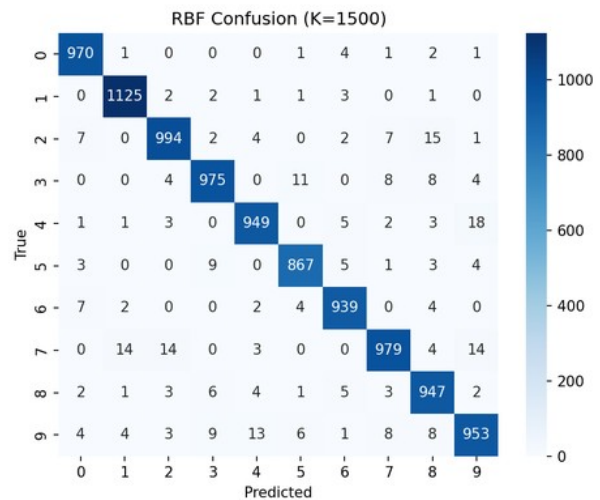


Figure 6: Confusion matrix for RBF (K = 1500)

With a larger number of centers, the RBF network significantly reduces several of the confusion patterns observed in the baseline model. The errors between 4 and 9, as well as 3 and 5, are notably reduced. This demonstrates that increasing model capacity leads to more refined decision boundaries and improved class separability.

Accuracy increases monotonically with K, as expected from the universal approximation properties of RBF networks.

2. Training Time vs K

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Figure 7: Training time as a function of the number of RBF centers K

Training time grows approximately linearly:

- 50 → 16 s
- 100 → 29 s
- 200 → 53 s
- 400 → 98 s
- **1500 → 436 s**

This highlights the trade-off between accuracy and computational cost.

3. Comparison of All Models

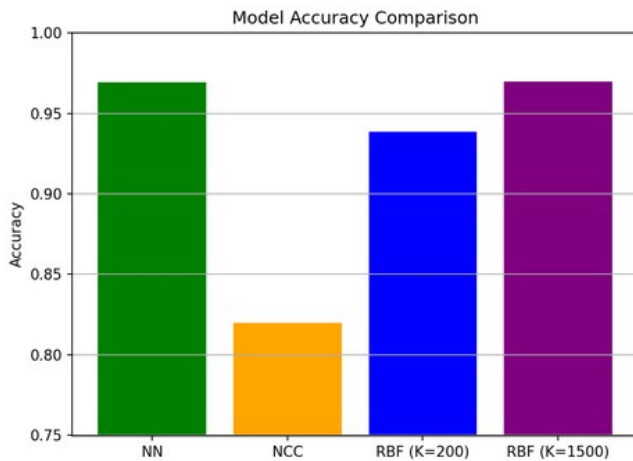


Figure 8: Accuracy comparison across classifiers

The bar chart compares the test accuracy of four models: Nearest Class Centroid (NCC), Nearest Neighbor (NN), RBF with $K = 200$, and RBF with $K = 1500$. NN and RBF ($K = 1500$) achieve the highest accuracy, while NCC performs significantly worse.

E. Center Initialization: K-means vs Random

Two strategies are compared:

- **K-means** consistently yields higher accuracy due to more representative centers.
- **Random sampling** is faster but less stable and typically 1–2% worse.

This confirms that center quality is crucial for RBF performance.

F. Bandwidth Estimation (σ)

Two methods are evaluated:

- **Median distance (global σ)**
- **Per-center σ (local bandwidths)**

Findings:

- Median σ is more stable and performs better for large K .
- Per-center σ can overfit when K is small, leading to inconsistent accuracy.

G. Ridge Regularization

Values tested: $\lambda \in \{0.001, 0.01, 0.1\}$

- $\lambda = 0.01$ provides the best balance between stability and accuracy.
- $\lambda = 0.001$ can lead to overfitting for large K .
- $\lambda = 0.1$ slightly reduces accuracy.

H. Effect of Random Seed

Seeds 42 and 123 are tested.

- Accuracy differences are minimal ($< 0.2\%$).
- Confirms robustness of the RBF pipeline.

I. Comparison of All Models

Model	Accuracy	Training Time
NCC	81.99%	0.09 s
NN	96.94%	2.4 s
RBF (K=200)	93.82%	60.7 s
RBF (K=400)	95.30%	98 s
RBF (K=1500)	96.98%	436 s

Key insight:

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The RBF network with **K = 1500** slightly **outperforms NN**, demonstrating that RBFs can achieve near state-of-the-art MNIST performance when sufficiently expressive.

V. Discussion

J. Qualitative Analysis

Examples of correct and incorrect predictions reveal:

- Most errors occur between visually similar digits (3/5, 4/9, 8/9).
- RBF tends to smooth decision boundaries, sometimes misclassifying ambiguous samples that NN handles correctly.
- However, RBF generalizes better than NN in some noisy cases.

K. Summary of Findings

- ❖ Increasing K improves accuracy but increases training time.
- ❖ PCA reduces dimensionality with negligible accuracy loss.
- ❖ K-means centers outperform random sampling.
- ❖ Median σ is more stable than per-center σ .
- ❖ RBF with large K can match or surpass NN accuracy.

The experimental results provide a detailed view of how Radial Basis Function (RBF) networks behave under different architectural and training configurations, and how they compare to classical non-parametric baselines on the MNIST digit classification task. Several important observations emerge from the analysis.

First, the number of hidden neurons K plays a central role in determining the expressive power of the RBF network. As K increases, the model becomes capable of representing more complex decision boundaries, leading to a consistent improvement in accuracy. This trend aligns with the theoretical interpretation of RBF networks as universal approximators. However, the computational cost grows substantially with larger K, both due to the K-means clustering step and the construction of the design matrix. The configuration with K=1500 achieves the highest accuracy (96.98%), slightly surpassing the 1-Nearest Neighbor baseline, but requires more than 400 seconds of training time. This highlights a clear trade-off between accuracy and efficiency.

Second, dimensionality reduction through PCA proves highly effective. Reducing the input dimensionality from 784 to 154 components (95% variance retention) leads to a dramatic reduction in training time with negligible impact on accuracy. This suggests that the discriminative structure of MNIST is largely preserved in a lower-dimensional subspace, and that RBF networks benefit from operating in a more compact and well-conditioned feature space. The stability of accuracy across different PCA variance levels further reinforces the robustness of the model to moderate feature compression.

Third, the choice of center initialization significantly influences performance. K-means clustering consistently outperforms random sampling, confirming that representative centers lead to smoother and more meaningful radial basis activations. Although random sampling is computationally cheaper, the accuracy gap (typically 1–2%) indicates that center quality is a critical factor in RBF design. Similarly, the bandwidth estimation method affects the smoothness of the decision boundaries. A global σ based on median distances provides stable performance across all values of K, while per-center σ can lead to overfitting when the number of centers is small.

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Fourth, the comparison with NN and NCC baselines highlights the strengths and limitations of each approach. NN achieves high accuracy with no training cost but suffers from high inference complexity and memory requirements. NCC, while extremely fast, performs poorly due to its oversimplified representation of each class. The RBF network occupies a middle ground: it requires more computation during training but offers competitive accuracy and efficient inference once trained. The fact that RBF with sufficiently large K can match or surpass NN accuracy demonstrates its potential as a flexible and powerful classifier.

Finally, the qualitative analysis of misclassifications reveals that the RBF network tends to struggle with digit pairs that are visually similar or ambiguous, such as 3/5, 4/9, and 8/9. These patterns are consistent with known MNIST challenges and appear across all models, indicating that the remaining errors are largely due to intrinsic dataset ambiguity rather than model deficiencies.

Overall, the results show that RBF networks remain a relevant and interpretable approach for multi-class classification. Their performance depends strongly on architectural choices, but with appropriate hyperparameters, they can achieve accuracy comparable to strong non-parametric baselines while offering advantages in interpretability and inference efficiency.

VI. CONCLUSIONS

This study examined the performance of Radial Basis Function (RBF) networks on the MNIST digit classification task and analyzed how key design choices affect accuracy and efficiency. The experiments showed that increasing the number of hidden neurons significantly improves performance, with the model reaching **96.98%** accuracy for $K=1500$, slightly surpassing the Nearest Neighbor baseline. However, this comes with a substantial increase in training time, highlighting a clear accuracy–efficiency trade-off.

Dimensionality reduction with PCA proved highly effective, reducing computational cost while preserving nearly all discriminative information. K-means center initialization and a global bandwidth estimate provided the most stable and reliable performance across configurations.

Overall, the results demonstrate that RBF networks remain a competitive and interpretable approach for multi-class classification. With appropriate hyperparameter choices, they can achieve strong accuracy while offering advantages in flexibility and inference efficiency.

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